



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

November 16, 2023

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 21030
(FIRST REVISION)

EXPIRATION DATE: 2027-10-31

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Mistras Group, Inc.
Princeton Junction, NJ
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the transportation in commerce of certain gases listed in paragraph 6 in certain composite overwrapped pressure vessels (COPVs) listed in paragraph 7 when Modal Acoustic Emission (MAE) examination is used for requalification in lieu of the hydrostatic pressure testing required in § 180.205 and the relevant special permits, as described in this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein.
 - b. The safety analyses performed in the development of this special permit only considered the hazards and risks associated with the transportation in commerce.
 - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 180.205(g) in that MAE is performed in place of the hydrostatic pressure test and internal visual inspection, as provided herein.
5. BASIS: This special permit is based on the application of Mistras Group, Inc. dated July 8, 2023, submitted in accordance with § 107.109.

Tracking Number: 2023065214

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Air, compressed (containing up to 39% by volume oxygen)	2.2	UN1002	N/A
Argon, compressed	2.2	UN1006	N/A
Bromotrifluoromethane <i>or</i> Refrigerant Gas, R 13B1	2.2	UN1009	N/A
Carbon dioxide	2.2	UN1013	N/A
Chlorodifluorobromomethane <i>or</i> Refrigerant Gas, R12B1	2.2	UN1974	N/A
Compressed gas, n.o.s.	2.2	UN1956	N/A
Helium, compressed	2.2	UN1046	N/A
Heptafluoropropane <i>or</i> Refrigerant Gas R 227	2.2	UN3296	N/A
Hydrogen, compressed	2.1	UN1049	N/A
Methane, compressed <i>or</i> Natural gas, compressed (<i>with high methane content</i>)	2.1	UN1971	N/A
Neon, compressed	2.2	UN1065	N/A
Oxygen, compressed	2.2	UN1072	N/A
Nitrogen, compressed	2.2	UN1066	N/A
Nitrous oxide	2.2	UN1070	N/A

7. SAFETY CONTROL MEASURES:

- a. PACKAGING: Packagings prescribed are composite overwrapped pressure vessels (COPV) that were manufactured in accordance with UN/ISO 11119-2 and

11119-3 and ISO 11515 and COPVs manufactured under DOT-SP 16524. These COPVs must be subjected to periodic inspection and testing (requalification) and marking as specified in §§ 180.205, 180.209, and 180.213 and as specified in each relevant special permit except that each COPV is examined by the Modal Acoustic Emission (MAE) method in lieu of hydrostatic pressure testing and internal visual inspection. Each COPV must be requalified in accordance with the procedures specified in this special permit and the application for this special permit on file with the Office of Hazardous Materials Safety (OHMS).

b. Requalification: Each COPV must be requalified once every five (5) years by an approved facility that has a DOT Requalifier Identification Number (RIN) from OHMS for requalification of the COPVs listed in paragraph 7.a. using Modal Acoustic Emission (MAE) testing in lieu of hydrostatic pressure testing and internal visual inspection in accordance with following:

(1) External Visual Examination: Each COPV must receive an external visual examination in accordance with CGA Pamphlet C-6.2 Standards for Visual Inspection and Requalification of Fiber Reinforced High Pressure Cylinders (COPVs) or ISO 11623 “Gas cylinders — Composite construction — Periodic inspection and testing”. Cylinders (COPVs) that do not meet C-6.2 acceptance criteria must be condemned.

(2) MAE Examination (Testing) procedure: MAE testing procedure and accept/reject criteria must be in accordance with the Modal Acoustic Emission (MAE) Examination Specification for Requalification of Composite Overwrapped Pressure Vessels (Cylinders and tubes) posted on the Pipeline and Hazardous Materials Safety Administration website, <https://www.phmsa.dot.gov/technical-resources/hazmat-technical-resources/technical-reports>, and the additional procedure and supporting documents provided by the applicant on file with OHMS.

c. MARKING: Retest markings must be applied on a label securely affixed to the COPV and over coated with epoxy, near the original test date. The retest markings must include the retest date, requalifier RIN, and “MAE” as the requalification test procedure. The marking of the RIN number on the COPV certifies compliance with all terms and conditions of this special permit.

d. REPORT: A report must be generated for each COPV listing all COPVs that were examined. The MAE reports must be on file at the test site and made available to a DOT representative when requested. The MAE reports must include the following:

(1) RIN.

(2) MAE technician’s name and certification level.

- (3) MAE equipment, model, and serial number.
- (4) COPV manufacture date, design type, serial number, special permit number, test pressure, and service pressure.
- (5) Test date.
- (6) MAE test pressure.
- (7) Decay rate B values.
- (8) The final value of each allowance factors (F1, F2, M1, M2) used to determine the accept/reject of the COPV.
- (9) The value of event energies exceeding the fiber bundle failure criteria.
- (10) The value of event energies exceeding the FRAE requirement.
- (11) Background energy oscillation pressure.
- (12) Visual inspection results (Pass/Fail).
- (13) Acceptance/rejection results (Pass/Fail).
- (14) Brief description for each condemned COPV.

e. PERSONNEL QUALIFICATION: Each person who performs retesting or who evaluates or certifies retest results must meet the following requirements:

- (1) Project Manager is the senior manager of Mistras Group, Inc. and is responsible for compliance with DOT regulations including this special permit. Additionally, the project manager must ensure that each operator and senior review technologist maintains the required certifications described herein.
- (2) The personnel responsible for performing COPV retesting under this special permit shall be qualified to an appropriate Level (Level I, II, or III) in Modal Acoustic Emission (MAE) testing in accordance with the American Society for Nondestructive Testing (ASNT) Recommended Practice SNT-TC-1A-1996 depending upon the assigned responsibility as described below:
 - (i) As a minimum, a Level II Operator must perform system startup, calibrate the system, and review and certify the test results when written acceptance and rejection criteria for COPVs has been provided by a Senior

Review Technologist. Based upon the written criteria, the Level II Operator may authorize COPVs that pass the retest to be marked in accordance with paragraph 7.c. of this special permit.

(ii) However, a person with Level I certification may perform a system startup, check calibration, and perform MAE under the direct guidance and supervision of a Senior Review Technologist or a Level II Operator, either of whom must be physically present at the test site to be able to observe examinations conducted under this special permit.

(3) Senior Review Technologist (SRT) is a person who provides written MAE procedure, supervisory training and examinations (Level I and II), technical guidance to operators, and reviews and verifies the retest results. The SRT must be capable of examining the waveforms (event by event), and the waveforms for each event must precisely correspond with the pressure and time data during the test. The SRT must prepare and submit the reports required in paragraph 7.d. and annually verify that the MAE program is being operated in accordance with the requirements of this special permit. An SRT must have a thorough understanding of the DOT Regulations (49 CFR) pertaining to the requalification and reuse of these COPVs authorized under this special permit and ASNT Recommended Practice SNT-TC-1A and possesses:

(i) A Level III certificate from ASNT in AE or UE with documented experience for a minimum of 2 years of experience in Non-Destructive Evaluation (NDE) of pressure vessels or pipelines using the Modal Acoustic Emission technique;

(ii) A Professional Engineer (PE) License with documented experience for a minimum of 2 years of experience in Non-Destructive Evaluation (NDE) of pressure vessels or pipelines using the Modal Acoustic Emission techniques; or

(iii) A PhD degree in Engineering/Physics with documented evidence of experience in Non-Destructive Evaluation (NDE) of pressure vessels or pipelines, using the Modal Acoustic Emission technique or research/thesis work and authoring/co-authoring of technical papers published, in recognized technical journals, in the field of the Modal Acoustic Emission testing method.

The most recent copies of certification (e.g., ASNT Level III or PE license) must be available for inspection at each requalification facility.

f. Operational Controls: Prior to requalification and marking a COPV under this special permit, Mistras Group, Inc. must complete the following:

- (1) The requalification facility must be inspected by a DOT approved Independent Inspection Agency (IIA). The IIA must witness and approve the MAE equipment, calibration and requalification procedure of COPVs listed in this special permit. Mistras Group, Inc. must apply for and obtain a Requalifier Identification Number (RIN) letter from OHMS; and
- (2) Standard Operating Procedure (SOP) for requalification of COPVs under this special permit that includes:
 - (i) Safety measures for pneumatic testing: location(s) and protective barrier (e.g. concrete bunker or remote location) that ensure the safety of the personnel during the COPV pressurization process when MAE method B (pneumatic) is used; and
 - (ii) Personnel safety procedures for the MAE technician(s) and others during pressurization, hold time, and depressurization.

8. SPECIAL PROVISIONS:

a. The designated SRT must review the MAE program annually. The designated SRT must submit a letter to OHMS verifying that the MAE program is meeting the terms and requirements of this special permit.

b. Annual Report: An annual report must be submitted to OHMS for each MAE requalification facility. The report includes the total number of COPVs that have been requalified under this special permit. The annual report must include:

- (1) Specification/Standard of each COPV that was requalified (e.g. ISO 11119-3).
- (2) Size (water volume), outside diameter (OD), design test pressure (P_h) and age of each COPV.
- (3) Value of the stress factors (F1, F2, M1, and M2) used during final analysis, accept or reject criteria for each COPV.
- (4) Defect type (i.e., fiber tow fracture (FTF), fretting emission (FE), fiber breakage, delamination cuts, abrasions, impacts, chemical damage, heat damage, etc.) that was the cause for condemnation of a COPV.
- (5) Location of the defect(s) that was the cause for condemnation of a COPV.

- c. A statement of qualifications, for each “qualified MAE tester” used under this special permit and information in support thereof, must be maintained by Mistras Group, Inc. The location of this statement, for each “qualified MAE tester”, must be identified to OHMS.
 - d. A current copy of the DOT special permit for the COPVs listed in this special permit must be maintained at each facility where retesting is performed.
 - e. Shippers (offerors) may use the COPVs specified and tested in accordance with the provisions of this special permit for the transportation in commerce of those hazardous materials specified herein, provided no modification or change is made to the COPVs. All terms of this special permit and other applicable requirements contained in 49 CFR Parts 100-185 must be met.
 - f. Transportation of Division 2.1 (flammable gases) and Division 2.3 (gases which are poisonous by inhalation) materials are not authorized aboard cargo vessel or aircraft unless specifically authorized in the Hazardous Materials Table (§ 172.101).
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, passenger-carrying aircraft, and cargo-only aircraft.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel or aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq.:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) — “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: Mark Toughiry/Casey Chambers